

Unified Dynamics as an Accommodation-Channel Extension of General Relativity

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April 14, 2026

Abstract

We isolate the gravitational branch of Unified Dynamics (UD), a packet-transport closure framework in which geometry is not treated as primitive but arises from transport, burden, and closure structure on a combinatorial substrate. Starting from a discrete packet action with curvature, burden, and orientation terms, we write an effective gravitational equation whose source decomposes into a standard GR sector plus four structured accommodation channels: closure-angle, packet-spacing, channel-slack, and branch occupation. We then state a relaxed coarse isotropic limit in which the accommodation terms are suppressed and the framework reduces to Einstein dynamics. The resulting architecture organizes possible departures from GR into a named correction hierarchy rather than an unconstrained parameter expansion, and it suggests falsifiable signatures for compact-object and gravitational-wave regimes. The purpose of this paper is to present that gravitational architecture in a controlled standalone form.

1 Introduction

General relativity remains the benchmark description of gravity in regimes where geometry can be treated as a smooth macroscopic object [1]. Many approaches to emergent or modified gravity begin by altering geometry directly or by adding new field content to an already geometric stage. Unified Dynamics (UD) starts earlier. It treats geometry as a coarse expression of transport, burden, and closure structure on a combinatorial packet substrate; see also the broader companion architecture for context [4].

In the broader UD program, the same substrate is also used to organize matter, gauge-sector, and cosmological branches. The present paper is narrower. It isolates only the gravitational branch and asks whether a packet-transport closure substrate supports an effective gravitational equation with a structured correction architecture and a controlled general-relativistic limit.

The answer proposed here is yes. The effective source decomposes into a standard GR sector plus four accommodation channels associated with closure-angle strain, packet-spacing strain, channel-slack strain, and branch-occupation strain. In the relaxed coarse isotropic limit these accommodation contributions are suppressed, and the framework reduces to Einstein dynamics.

This paper does not attempt to derive the full matter, gauge, or cosmological sectors of the broader UD program; it isolates only the gravitational branch and its effective limit structure. A central task of the paper is to show how the native stress terms of the packet action coarse-grain into the accommodation-channel source structure used by the effective gravitational equation.

2 Packet Substrate, Transport, and Action

The primitive arena is a combinatorial packet complex

$$K = (V, E, F, T, \dots),$$

with packet sites V , transport links E , triangular transport faces F , and tetrahedral closure cells T . Each site carries a local packet state

$$X_i = (\rho_i, \theta_i, \chi_i, \sigma_i, b_i), \quad (1)$$

where ρ_i is local burden or loading, θ_i is transport phase or orientation, χ_i is closure-branch state, σ_i is framing class, and b_i is boundary or admissibility exposure. The primitive substrate and its loop-based transport geometry are illustrated schematically in Fig. 1.

For the gravitational branch, the most important components are burden, transport orientation, and closure structure. Burden determines how much unresolved loading is carried locally; transport orientation determines how neighboring packets exchange or mismatch phase; and closure structure determines whether transport loops remain compatible or accumulate geometric stress. In the gravitational branch, burden is interpreted as unresolved substrate loading whose coarse redistribution contributes to effective stress-energy.

The discrete packet action is

$$S_{\text{pkt}} = \sum_C \kappa_C \text{Tr}(I - U_C) + \sum_i \lambda_B (\rho_i - \rho_0)^2 + \sum_{\langle ij \rangle} \eta_O (\theta_i - \theta_j)^2, \quad (2)$$

where κ_C is the curvature coupling, λ_B is the burden-relaxation cost, and η_O is the orientation-mismatch cost. The first term penalizes holonomy mismatch around closed transport loops, the second penalizes deviation from equilibrium burden, and the third penalizes unresolved orientation gradients across neighboring packet sites.

Transport along an oriented path $\mathcal{P}$ is represented by

$$U(\mathcal{P}) = \prod_{(ij) \in \mathcal{P}} U_{ij}, \quad (3)$$

while simplicial curvature is captured by the deficit-angle analog

$$\delta_e = 2\pi - \sum_{\tau \supset e} \Theta_\tau, \quad (4)$$

in the spirit of discrete geometric curvature constructions [7, 2]. Figure 1 emphasizes the two gravity-relevant primitive objects of the present paper: packet-local state and transport composed around a closed loop.

2.1 From Packet Action to Effective Accommodation Source

The effective source structure developed below is the coarse-grained image of the three native terms in the packet action Eq. (2). At discrete level, the action contains three irreducible stress-generating mechanisms:

(i) *Holonomy mismatch*:

$$\sum_C \kappa_C \text{Tr}(I - U_C),$$

which penalizes unresolved transport curvature around closed loops.

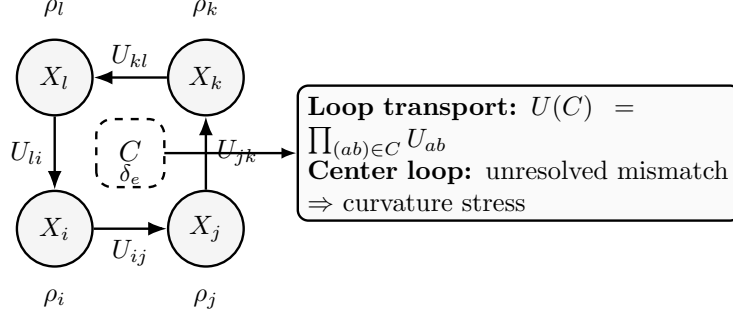


Figure 1: Packet substrate and loop holonomy. Packet sites carry local states X_i , oriented links carry transport operators U_{ij} , and transport composed around a closed loop C defines the local holonomy. In the gravitational branch, unresolved loop mismatch is the discrete source of curvature-sensitive stress.

(ii) *Burden deviation:*

$$\sum_i \lambda_B (\rho_i - \rho_0)^2,$$

which penalizes local departure from equilibrium loading.

(iii) *Orientation mismatch:*

$$\sum_{\langle ij \rangle} \eta_O (\theta_i - \theta_j)^2,$$

which penalizes unresolved transport-angle gradients across neighboring packet sites.

Under coarse-graining, these terms do not remain as independent microscopic penalties. Instead, they appear as effective substrate stresses organized by the dominant non-isotropic ways in which closure-capable packet structure can deviate from relaxed equilibrium.

First, holonomy mismatch contributes closure-sensitive stress. At coarse level this appears as the closure-angle accommodation channel $\Delta T_{\mu\nu}^{(\theta)}$, since unresolved loop mismatch is registered macroscopically as angular closure strain. When such mismatch is directionally biased by persistent branch loading, the same sector also contributes to branch-sensitive corrections, represented at effective level by $\Delta T_{\mu\nu}^{(b)}$.

Second, burden deviation contributes storage and compression stress. Its most direct coarse image is the packet-spacing accommodation channel $\Delta T_{\mu\nu}^{(a)}$, because excess or deficit local burden changes the effective spacing and packing profile of the substrate relative to relaxed equilibrium.

Third, orientation mismatch contributes anisotropic transport stress. At coarse level this appears most naturally as the channel-slack accommodation channel $\Delta T_{\mu\nu}^{(c)}$, since unresolved directional transport gradients manifest as imbalance in transport-capacity utilization across the substrate.

Cross-couplings arise because the underlying packet action is not separable under coarse evolution; burden, orientation, and closure mismatch can activate one another away from the isotropic relaxation limit. This motivates the additional effective term $\Delta T_{\mu\nu}^{(\text{cross})}$.

Accordingly, the effective gravitational equation below should be read as the leading coarse-grained organization of packet-substrate stress rather than as an independent continuum ansatz imposed from outside the discrete theory.

3 Effective Gravitational Equation

In a relaxed coarse isotropic limit, the coarse-grained substrate stresses generated by Eq. (2) organize into an effective gravitational equation while allowing structured departures away from that limit. The source architecture of the gravitational branch is summarized schematically in Fig. 2.

Relaxed isotropic reference state. The relaxed isotropic reference state is the coarse regime in which mean packet spacing is homogeneous, closure-angle strain averages to zero, transport utilization is directionally balanced, and branch occupation is exchangeable under spatial averaging. The accommodation channels classify departures from this reference state.

The effective field equation is written as

$$G_{\mu\nu} = \kappa_G \left[T_{\mu\nu}^{(\text{GR})} + \Delta T_{\mu\nu}^{(\theta)} + \Delta T_{\mu\nu}^{(a)} + \Delta T_{\mu\nu}^{(c)} + \Delta T_{\mu\nu}^{(b)} + \Delta T_{\mu\nu}^{(\text{cross})} \right], \quad (5)$$

where $\kappa_G = 8\pi G$ and

$$T_{\mu\nu}^{(\text{GR})} = (\rho_m + p_m)u_\mu u_\nu + p_m g_{\mu\nu}$$

is the standard GR source tensor [1].

The correction terms are not introduced as an unconstrained beyond-GR expansion. Instead, they are organized into four accommodation channels determined by native substrate strains.

Closure-angle channel. $\Delta T_{\mu\nu}^{(\theta)}$ records deviations in angular closure coherence from isotropic equilibrium.

Packet-spacing channel. $\Delta T_{\mu\nu}^{(a)}$ records deviations in effective inter-packet separation from homogeneous equilibrium.

Channel-slack channel. $\Delta T_{\mu\nu}^{(c)}$ records imbalance in transport-capacity utilization relative to a balanced-load regime.

Branch channel. $\Delta T_{\mu\nu}^{(b)}$ records deviations in directional branch occupation from exchangeable equilibrium.

These four channels are intended as the minimal coarse-grained decomposition of non-isotropic substrate strain in the present framework. Minimality here means that each channel corresponds to a distinct lowest-order departure from the relaxed isotropic reference state: angular closure distortion, spacing distortion, transport-capacity imbalance, and directional branch asymmetry. As shown in Fig. 2, the four accommodation channels are interpreted as the lowest-order departures from the relaxed isotropic reference state, while cross-couplings encode mixed activation away from that limit.

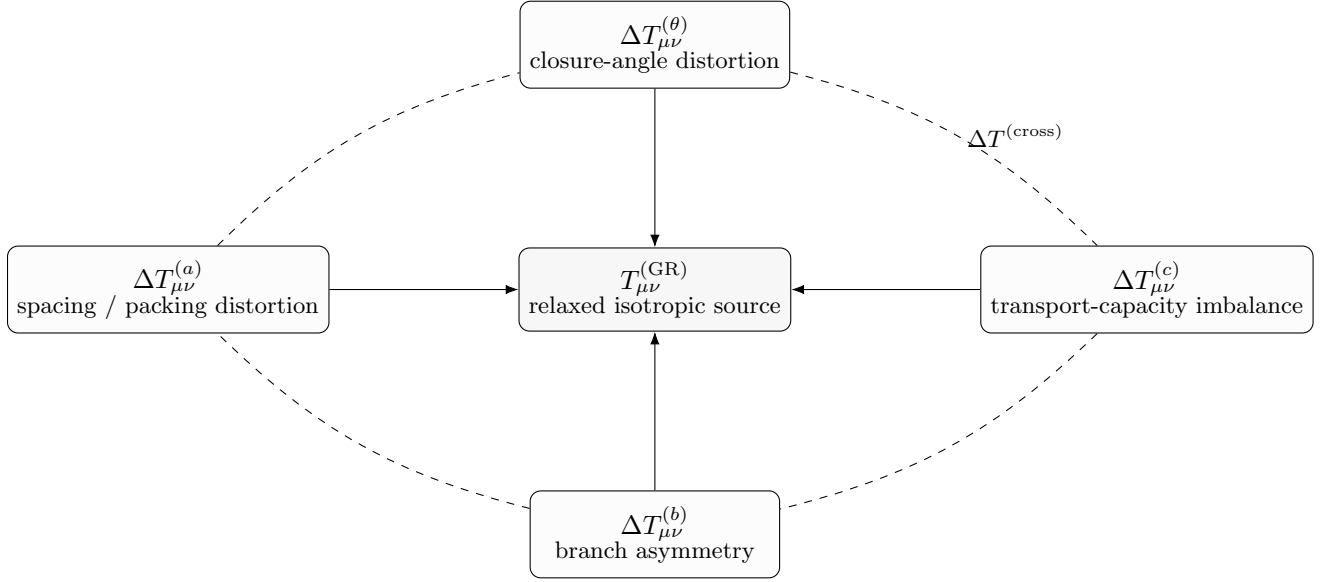
The corresponding elastic storage density is

$$W_{\text{el}} = A_a \varepsilon_a^2 + A_\theta \varepsilon_\theta^2 + A_c \varepsilon_c^2 + A_b \varepsilon_b^2 + W_{\text{el}}^{(\text{cross})}, \quad (6)$$

where the ε 's measure fractional deviations of the native channel observables from isotropic equilibrium, and $W_{\text{el}}^{(\text{cross})}$ collects cross-coupling terms such as $A_{(a\theta)} \varepsilon_a \varepsilon_\theta$ and $A_{(\theta c)} \varepsilon_\theta \varepsilon_c$.

The physical meaning of accommodation is that part of the substrate burden can be stored in channel-specific deformation before reappearing as coarse geometric response. In this way, the framework organizes modified gravitational response into structured residual families rather than a flat correction zoo.

Accommodation-channel decomposition of the effective source



Lowest-order departures from the relaxed isotropic reference state

Figure 2: Accommodation-channel decomposition of the effective gravitational source. Relative to the relaxed isotropic reference state, non-isotropic substrate strain is organized into four lowest-order accommodation channels: closure-angle, packet-spacing, channel-slack, and branch occupation. Cross-couplings encode mixed activation away from the isotropic relaxation limit.

4 GR Limit of the Accommodation Architecture

The effective source Eq. (5) reduces to Einstein dynamics in a relaxed coarse isotropic regime.

Proposition 1 (GR limit of the accommodation architecture). *Suppose that the following conditions hold at coarse level:*

- (i) *anisotropic transport flux satisfies $q_\mu \rightarrow 0$,*
- (ii) *anisotropic stress satisfies $\pi_{\mu\nu} \rightarrow 0$,*
- (iii) *branch occupation becomes directionally exchangeable,*
- (iv) *coherence orientations decorrelate at large scale,*
- (v) *no persistent direction-locking field survives coarse averaging.*

Under these isotropization and decorrelation assumptions, the accommodation contributions are suppressed in the coarse-grained source, and the effective field equation reduces to

$$G_{\mu\nu} = \kappa_G T_{\mu\nu}^{(GR)}.$$

In this sense, general relativity appears as the relaxed coarse isotropic limit of the accommodation architecture: not as a primitive starting point of the framework, but as the limiting output of a specific coarse regime.

5 Curvature-Coupling Normalization

The present gravitational branch uses the structural normalization

$$\kappa_C = \frac{3}{4\pi}, \quad (7)$$

motivated by the discrete curvature budget of tetrahedral packet geometry [7]. At synthesis level, this quantity functions as the working curvature coupling of the packet action Eq. (2).

In the present paper, κ_C is used as the working structural normalization of the curvature term; its fuller topological derivation is not required for the main accommodation-channel argument. What matters here is that the curvature term in the packet action is normalized by a definite structural scale rather than treated as an arbitrary free correction coefficient. Supplementary derivational notes for the gravitational branch may be provided separately [6].

A preliminary Regge-style scaling result from the broader Gap 5 analysis pipeline is shown in Appendix Fig. 4; in the present paper it is used only as methodological support for the possibility of coarse geometric scaling on the packet substrate.

6 Regimes, Observables, and Falsifiers

The accommodation architecture suggests a regime ordering for gravitational behavior. The regime ordering of the accommodation architecture, together with its relaxation toward the GR limit, is shown schematically in Fig. 3.

Ordinary weak-field regime. When all accommodation strains are small and isotropic relaxation is effective, the correction channels remain subleading and the Einstein limit is an excellent approximation.

Compact-object regime. Near strongly loaded or rapidly evolving systems, multiple accommodation channels may become active simultaneously. In such cases, departures from GR should appear not as arbitrary anomalies but as structured mixtures of closure-angle, spacing, channel-slack, and branch effects.

Organized branch-active regime. In systems with persistent directional transport or branch asymmetry, branch-channel effects may remain visible beyond the simplest isotropic picture.

Cross-coupled regime. When several accommodation strains are simultaneously large, the cross terms in Eq. (6) become important and the correction hierarchy cannot be treated channel-by-channel in isolation.

Figure 3 makes clear that the accommodation channels are not a static taxonomy: their visibility depends on how far the substrate remains from relaxed isotropic balance.

At hypothesis level, the accommodation channels are expected to bias different classes of residual behavior in compact-object and gravitational-wave settings [3]. The closure-angle channel is expected to modify closure-sensitive tidal response; the packet-spacing channel modifies effective radial structure; the channel-slack channel affects transport-capacity asymmetry; and the branch channel introduces directional occupation effects.

The gravitational branch fails if any of the following conditions are robustly established:

Regime ordering and relaxation toward the GR limit

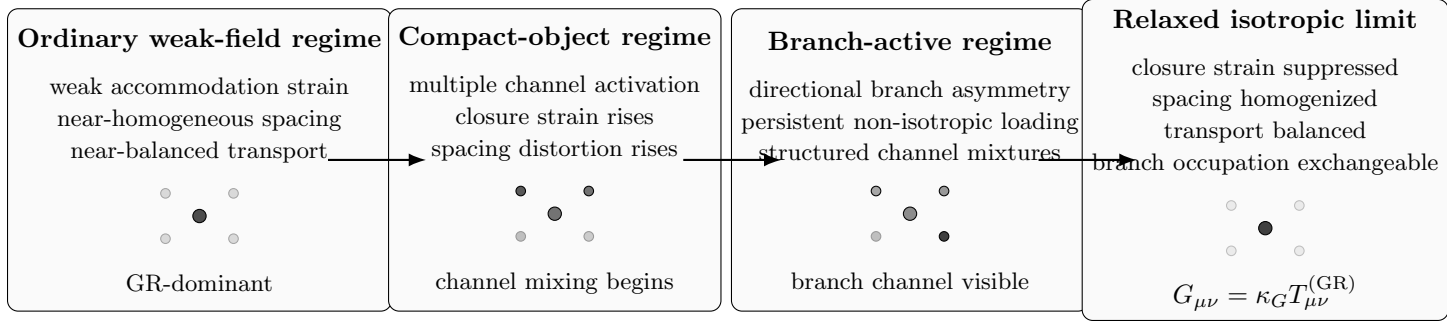


Figure 3: Regime ordering and the GR limit of the accommodation architecture. Away from isotropic relaxation, gravitational response may organize into structured accommodation-channel mixtures. As spacing, closure strain, transport utilization, and branch occupation relax toward isotropic balance, the accommodation contributions are suppressed and the effective source reduces to the GR sector.

- (1) large clean non-GR deviations are observed in ordinary isotropic weak-field matter where accommodation effects should be negligible;
- (2) no controlled coarse isotropic limit exists in which the accommodation terms vanish and Einstein dynamics are recovered;
- (3) compact-object data decisively excludes the possibility of organizing residual structure into the accommodation-channel hierarchy;
- (4) the correction architecture requires ad hoc unconstrained channel additions rather than the four native channels and their cross-couplings.

7 Conclusion

We have isolated the gravitational branch of Unified Dynamics and written it as an effective gravitational equation sourced by a standard GR sector plus four structured accommodation channels. Starting from a packet-transport closure substrate and its associated packet action, the framework organizes modified gravitational response into structured channel families rather than an unconstrained correction search.

Under relaxed coarse isotropic conditions, the accommodation terms are suppressed and the framework reduces to Einstein dynamics. Away from that limit, the theory predicts organized residual behavior associated with closure-angle, packet-spacing, channel-slack, and branch-occupation strain.

The next test of the framework is operational rather than rhetorical: whether the packet substrate supports the stated source structure under executable dynamics, and whether gravitational data can distinguish the proposed accommodation families from ordinary GR residuals.

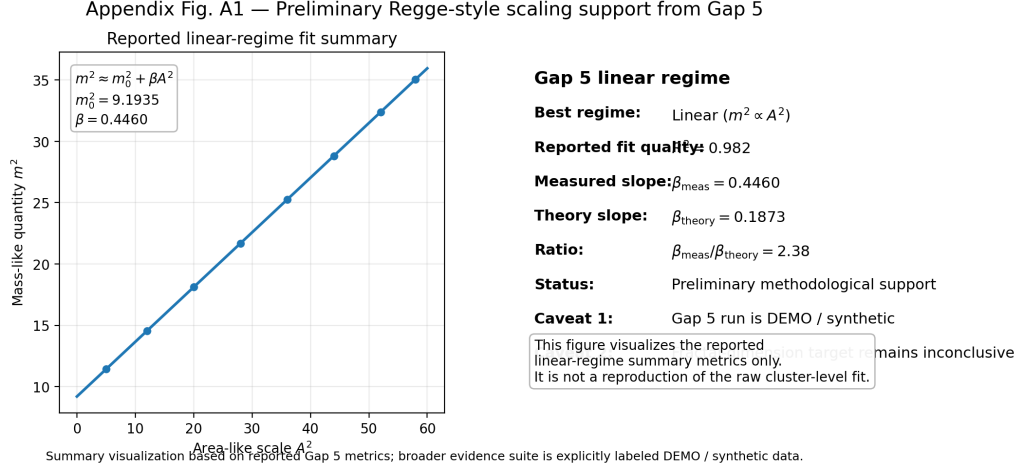


Figure 4: Preliminary Regge-style scaling evidence from the Gap 5 analysis pipeline. The linear-regime fit between the extracted mass-like quantity and area-like scale shows strong internal linearity in the current test run ($R^2 \approx 0.982$). In the context of the gravitational branch, this panel is used only as methodological support for the claim that the packet substrate can admit a meaningful coarse geometric scaling structure. It should not be read as definitive confirmation of the full geometric program, since the current Gap 5 report is explicitly based on demo synthetic data and the broader fractal-dimension target remains inconclusive.

A Preliminary Geometric-Scaling Support

The gravitational branch developed in the main text does not depend on the full closure of the broader geometric evidence program. However, it is useful to note that the current Gap 5 analysis pipeline already shows a strong linear Regge-style scaling relation in its linear regime. In the present paper, this result is interpreted conservatively: it provides preliminary methodological support for the possibility that the packet substrate admits a meaningful coarse geometric scaling structure. At the same time, the current Gap 5 report explicitly labels the run as demo synthetic data and classifies the primary fractal-dimension target as inconclusive [5]. For that reason, Fig. 4 is included only as supporting context rather than as a load-bearing premise of the accommodation-channel gravity argument.

References

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